ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 3.01: Indices, Surds & Polynomials

Q1. The variables x and y are related by the equation y = 2x^3 - 5x. Given that x is increasing at a constant rate of 3text{ units per second}, find the rate of change of y with respect to time at the instant when x = 2.

[x] A: 57text{ units per second}

[ ] B: 19text{ units per second}

[ ] C: frac{19}{3}text{ units per second}

[ ] D: 21text{ units per second}

Solution: To find the rate of change of y with respect to time, apply the chain rule for connected rates of change: frac{mathrm{d}y}{mathrm{d}t} = frac{mathrm{d}y}{mathrm{d}x} times frac{mathrm{d}x}{mathrm{d}t} 1. Differentiate y with respect to x: frac{mathrm{d}y}{mathrm{d}x} = frac{mathrm{d}}{mathrm{d}x}left2x^3 - 5xright = 6x^2 - 5 2. Evaluate frac{mathrm{d}y}{mathrm{d}x} at x = 2: left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=2} = 62^2 - 5 = 64 - 5 = 24 - 5 = 19 3. Multiply by frac{mathrm{d}x}{mathrm{d}t} = 3: frac{mathrm{d}y}{mathrm{d}t} = 19 times 3 = 57text{ units per second} Common errors: - 19 corresponds to finding frac{mathrm{d}y}{mathrm{d}x} only and neglecting frac{mathrm{d}x}{mathrm{d}t}. - frac{19}{3} corresponds to inverting the chain rule dividing instead of multiplying by frac{mathrm{d}x}{mathrm{d}t}. - 21 results from incorrectly differentiating 2x^3 as 6x instead of 6x^2. [M1][A1]

Q2. Consider the cubic polynomial fx = x^3 - 5x^2 + 10x - 20 . Find the remainder f2 upon division by x - 2 .

[ ] A: -10

[ ] B: -7

[x] C: -12

[ ] D: -22

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** By the Remainder Theorem, f2 = 2^3 - 52^2 + 102 - 20 = -12 . **[M1]** - **Step 3 Final Accuracy:** Verified exact result: -12. **[A1]**

Q3. The diagram shows part of the curve y = 6x - x^2 , which intersects the x -axis at 0, 0 and 6, 0 . Find the area of the region enclosed between the curve and the x -axis.

[ ] A: 27

[x] B: 36

[ ] C: 54

[ ] D: 72

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** To find the area enclosed between the curve and the x -axis, integrate y with respect to x from x = 0 to x = 6 : text{Area} = int_{0}^{6} 6x - x^2 , mathrm{d}x Integrate term by term: int 6x - x^2 , mathrm{d}x = left[frac{6x^2}{2} - frac{x^3}{3} right]_{0}^{6} = left[3x^2 - frac{x^3}{3} right]_{0}^{6} Substitute the limits: left 36^2 - frac{6^3}{3} right - 0 = 336 - frac{216}{3} = 108 - 72 = 36 Common errors: - Estimating the region as a triangle with base 6 and maximum height 9 yields frac{1}{2} times 6 times 9 = 27 . - Taking the bounding rectangle gives 6 times 9 = 54 . - Integrating 6x as 6x^2 without dividing by 2 gives 216 - 72 = 144 , or subtracting 108 - 36 = 72 . **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 36. **[A1]**

Q4. Solve the logarithmic equation log_25x + 6 - log_2x = 3 .

[x] A: x = 2

[ ] B: x = 6

[ ] C: x = frac{1}{2}

[ ] D: x = -3

Solution: 1. Apply the logarithm quotient law log_aA - log_aB = log_aleftfrac{A}{B}right : log_2leftfrac{5x + 6}{x}right = 3 2. Convert the equation from logarithmic to exponential form: frac{5x + 6}{x} = 2^3 = 8 3. Multiply through by x noting x > 0 : 5x + 6 = 8x implies 3x = 6 implies x = 2 4. Checking the domain: Both 52 + 6 = 16 > 0 and 2 > 0 , so x = 2 is a valid solution. Common misconceptions: - Calculating 2^3 = 6 instead of 8 yields 5x + 6 = 6x implies x = 6 . - Incorrectly subtracting the arguments directly as log_25x+6 - x = 3 leads to 4x + 6 = 8 implies x = frac{1}{2} . - Forgetting to exponentiate the base and equating frac{5x+6}{x} = 3 yields x = -3 , which also violates the logarithm domain. [M1][A1]

Q5. Find the value of the integer n ge 3 that satisfies the equation {}^{n}mathrm{P}_{3} = 10 times {}^{n}mathrm{C}_{2}

[x] A: n = 7

[ ] B: n = 12

[ ] C: n = 5

[ ] D: n = 8

Solution: Using the algebraic definitions of permutations and combinations: {}^{n}mathrm{P}_{3} = frac{n!}{n - 3!} = nn - 1n - 2 {}^{n}mathrm{C}_{2} = frac{n!}{2!n - 2!} = frac{nn - 1}{2} Substitute these into the given equation: nn - 1n - 2 = 10 times frac{nn - 1}{2} nn - 1n - 2 = 5nn - 1 Since n ge 3 , we have nn - 1 ne 0 . Dividing both sides by nn - 1 gives: n - 2 = 5 implies n = 7 Common misconceptions: - Forgetting the 2! denominator in {}^{n}mathrm{C}_{2} treating it as {}^{n}mathrm{P}_{2} gives n - 2 = 10 implies n = 12 . - Dividing by nn - 1 and erroneously writing n = 5 instead of n - 2 = 5 gives n = 5 . - Confusing the factorial simplification to set n - 3 = 5 leads to n = 8 . [M1][A1]

Q6. Find the remainder when the polynomial fx = 4x^3 - 6x^2 + 3x - 1 is divided by 2x - 1 .

[x] A: -frac{1}{2}

[ ] B: frac{1}{2}

[ ] C: -frac{9}{2}

[ ] D: 0

Solution: By the Remainder Theorem, when a polynomial fx is divided by a linear factor of the form ax - b , the remainder is fleftfrac{b}{a}right . Here, the divisor is 2x - 1 , so we set 2x - 1 = 0 implies x = frac{1}{2} . Substitute x = frac{1}{2} into fx : fleftfrac{1}{2}right = 4leftfrac{1}{2}right^3 - 6leftfrac{1}{2}right^2 + 3leftfrac{1}{2}right - 1 = 4leftfrac{1}{8}right - 6leftfrac{1}{4}right + frac{3}{2} - 1 = frac{1}{2} - frac{3}{2} + frac{3}{2} - 1 = -frac{1}{2} Therefore, the remainder is -frac{1}{2} . Common misconceptions: - Solving 2x - 1 = 0 incorrectly as x = -frac{1}{2} gives fleft-frac{1}{2}right = -frac{9}{2} . - Arithmetic sign errors when simplifying the terms often lead to frac{1}{2} or 0 . [M1][A1]

Unit 3.01: Indices, Surds & Polynomials

Q7. In triangle ABC , sides of lengths 32 and 10 meet at an included angle of 60^circ opposite side x . Find the squared length f = x^2 by the Law of Cosines.

[x] A: 804

[ ] B: 803

[ ] C: 820

[ ] D: 809

Solution: **Mark Scheme CAIE / Edexcel Grading Steps:** - **Step 1 Method & Formulation:** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]** - **Step 2 Derivation & Calculation:** By the Law of Cosines, x^2 = 32^2 + 10^2 - 23210cos60^circ = 804 . **[M1]** - **Step 3 Final Accuracy:** Verified exact result: 804. **[A1]**

Q8. The position vectors of points A and B relative to an origin O are vec{OA} = 2mathbf{i} - 3mathbf{j} and vec{OB} = 7mathbf{i} + 9mathbf{j}. Find the unit vector in the direction of vec{AB}.

[x] A: frac{1}{13}5mathbf{i} + 12mathbf{j}

[ ] B: frac{1}{13}-5mathbf{i} - 12mathbf{j}

[ ] C: frac{1}{sqrt{61}}5mathbf{i} + 6mathb

[ ] D: frac{1}{15}9mathbf{i} + 12mathbf{j}

Solution: First, find the displacement vector vec{AB}: vec{AB} = vec{OB} - vec{OA} = 7mathbf{i} + 9mathbf{j} - 2mathbf{i} - 3mathbf{j} = 7 - 2mathbf{i} + 9 - -3mathbf{j} = 5mathbf{i} + 12mathbf{j}. Next, calculate the magnitude of vec{AB}: |vec{AB}| = sqrt{5^2 + 12^2} = sqrt{25 + 144} = sqrt{169} = 13. The unit vector in the direction of vec{AB} is: frac{vec{AB}}{|vec{AB}|} = frac{1}{13}5mathbf{i} + 12mathbf{j}. - Choice B gives the unit vector for vec{BA} = vec{OA} - vec{OB}, which is in the opposite direction. - Choice C results from an arithmetic sign error in the mathbf{j}-component: 9 - 3 = 6 instead of 9 - -3 = 12. - Choice D results from incorrectly adding the position vectors vec{OA} + vec{OB}. [M1][A1]